PART 1

1. **[3 points]** Answer the following questions:

   (a) What are the design goals that a *good* MapReduce algorithm should target, relatively to: number of rounds R, local space M_L, and aggregate space M_A?

   (b) Any sequential algorithm can be straightforwardly transformed into a 1-round MapReduce algorithm which uses one reducer: explain why this does not yield a *good* MapReduce algorithm.

2. **[3 points]** Refer to Round 2 of the algorithm for k-center with z outliers implemented in Homework 3: briefly describe what the round does, and how the number of points it processes is affected by increasing the parameter L (number of partitions).

3. **[4 points]** Let $S = \{c_1, c_2, \ldots, c_k\}$ be the solution computed by Farthest-First Traversal on a pointset P, and let $\Phi_{\text{kcenter}}(P, S)$ be its objective function value. Define q as the point of P farthest from the centers. Using, without proof, the fact that $d(q, S)$ is less than or equal to the distance between *any two points* in $\{c_1, c_2, \ldots, c_k, q\}$, show that $\Phi_{\text{kcenter}}(P, S) \leq 2\Phi^{\text{opt}}_{\text{kcenter}}(P, k)$.

4. **[3 points]** Consider the computation of the distinct elements of a stream $\Sigma = x_1, x_2, \ldots x_n$. Explain how the Probabilistic Counting algorithm processes an element x_j and how it derives the final estimate.

PART 2

1. **[7 points]** Let P be a set of N *temperatures* measured by several sensors. Each element of P is a pair $(i, (s_i, temp_i))$, where $i \in [0, N-1]$ is a distinct integer key, s_i is a sensor, and $temp_i$ is a temperature. No assumptions are made on the number of pairs in P from the same sensor.

   (a) Let $K > 0$ be an integer constant. Design a 2-round MapReduce algorithm that, given P, returns a subset T of values $(s_i, temp_i)$ from pairs in P, selected as follows. For every sensor s, if in P there are $\leq K$ pairs $(i, (s_i, temp_i))$ with $s_i = s$, then T must contain the values of all these pairs, otherwise T must contain the values of exactly K of them, arbitrarily selected.

   (b) Analyze the local and aggregate space required by your algorithm. For full score, your algorithm must require $o(N)$ local space and linear aggregate space.

   (c) How does the local and aggregate space of your algorithm change if $K = \log_2 N$?

2. **[6 points]** Let A_1 and A_2 be two n-bit arrays used as Bloom filters to assess the membership for two sets S_1 and S_2, respectively, from the same universe U. Suppose that both arrays have been initialized by applying the same set of k hash functions $h_1, h_2, \ldots, h_k$, to S_1 (in case of A_1) and S_2 (in case of A_2).

   (a) Describe how to combine A_1 and A_2 into a new n-bit array A and how to use it to assess membership for $S_1 \cup S_2$, with no false negatives (i.e., elements of $S_1 \cup S_2$ erroneously claimed not to be in $S_1 \cup S_2$).

   (b) Give an estimate of the probability that $A[i] = 0$, for any $i \in [0, n-1]$.